

# Topological 0-Matrix (Substrate) Mechanics.

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*This constitutes a proposal for the unification of General Relativity and the Standard Model. Drawing upon data accumulated through decades of observations and experimental research, TSM (Topological Substrate Mechanics) synthesizes these two pillars of modern physics into a single, coherent framework. The formulation of this mathematical formalism would not have been possible without building upon the seminal works of numerous distinguished theoretical physicists and mathematicians. TSM integrates their key achievements, transforming the existing rigorous conceptual apparatus into a unified language for describing reality, transcending traditional divisions.*

## Chapter 0: Axioms of the 0-Matrix, or the Mechanical "Engine" of Reality

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Before proceeding to the formal mathematical apparatus, we must establish the foundation upon which the entire observable Universe rests. For decades, theoretical physics has erred by introducing abstract equations while failing to answer the most fundamental question: what is the world mechanically constructed from? This chapter strips reality of its mathematical fog and returns to the basics of kinematics. Its primary objective is to reset the reader's research intuition and liberate the inferential process from the dogmas deeply rooted in modern physics, which conflate mathematical abstraction with physical reality.

### 0.1. Rejection of the Non-Physical Void

One of the greatest challenges of modern physics is explaining the nature of the vacuum. Intuition suggests that a wave is always a perturbation of a medium: a stone thrown into water creates propagating ripples, and sound requires air to reach our ears. Classical physics has long accepted this as self-evident; however, when transitioning to light or radio waves, we suddenly accept that they transport energy through an absolute void. This is a logical inconsistency that can no longer be ignored.

We reject this dogma by introducing the first axiom: space is not empty. It is filled with fundamental elements of matter—0-particles. Their dimensions are of a sub-Planckian order, rendering them entirely undetectable as "objects" by our current technologies, yet they constitute a unified, dense substrate of reality.

These particles possess no charges or complex internal structure. As the simplest, indivisible units of matter, they are characterized by an identical, elementary mass and—critically—**a strictly defined, non-zero intrinsic volume** ( $V_0 > 0$ ). We thereby exclude from the theory the abstract concept of a zero-dimensional mathematical "point mass". In TSM, every constituent element of space possesses a physical dimension. In this model, every energetic state of the matrix reduces to pure kinetics—momentum resulting from the continuous collisions of the spherical representations of these particles. Everything we observe as physical phenomena is merely a macroscopic echo of these microscopic, ceaseless collisions.

### 0.2. Elasticity from Pure Kinetics

Traditional physics perceives the elasticity of materials through the prism of "invisible springs"—forces that bind particles together and resist alterations to their mutual positioning. In our model, we reject this description because it requires assuming the existence of forces within a structure prior to explaining the

nature of force itself. We do not introduce any additional force fields or attractive interactions. Instead, all elasticity is derived exclusively from the dynamics of 0-particles and their momentum.

Let us imagine a 0-particle confined within a closed volume (an oscillation sphere). In an equilibrium state, the particle executes oscillations, striking the boundaries of its sphere with a specific frequency. What occurs when an external force begins to compress this space? The mean free path of the particle is shortened. Maintaining a constant momentum, the return time between collisions decreases drastically, and the frequency of impacts against the chamber boundaries escalates exponentially.

For a macroscopic observer, this rapid increase in collision frequency is equivalent to a rise in pressure. This creates an elastic structure that resists compression—not due to "hidden forces," but through pure geometry and kinetics. This law constitutes the bedrock of our theory: elasticity here is an emergent property of collision density. The more we reduce the space, the more powerful the resistance becomes, which is perceived at the macro scale as the ideal elasticity of the medium. This mechanism, however, possesses a hard geometric limit: compression cannot proceed indefinitely, as it encounters an insurmountable threshold of random close packing of spheres ( $\phi_{\max}$ ), at which point the 0-particles come into contact with their physical boundaries.

### 0.3. The 4D Pressure Vessel: Finiteness as a Source of Stress

If the Universe were infinite, the momentum of 0-particles would dissipate irretrievably, and collisions would cease. For reality to exist as a stable structure, it must satisfy the condition of finiteness.

Our 4-dimensional space functions in this model like a gigantic, closed vessel. Because the 0-particles have nowhere to escape, they are forced into continuous collisions. This permanent, kinematic pressure ensures that space is never relaxed; it remains in a state of perpetual stress, thereby capable of instantaneously transmitting vibrations. In this view, the Universe is not filled with energy—it *is* that energy, confined within a finite geometry.

### 0.4. The State of Relaxation: Self-Regulation of 0-Particles

Every 0-particle tends toward a state of relaxation—a point where the pressure of impacts from one side perfectly balances the pressure from neighboring 0-particles. On the micro scale, this implies that a local densification of 0-particles immediately triggers a sharp increase in their kinetic activity, which the system treats as an "excess of pressure."

This phenomenon is analogous to the equalization of pressure in a gas vessel: particles in a region of higher collision density inevitably transfer their momentum further to regions of lower density, striving to uniformize the pressure throughout the vessel. The system does not allow an excess of energy to remain localized; it immediately dissipates it through a cascade of collisions with its neighbors. This mechanism ensures that no local deformation can remain permanent in the low-energy regime. Should any factor disrupt this arrangement, a pressure wave will propagate through the entire system until all elements return to a state of equilibrium. This drive toward the continuous equalization of stresses is termed "relaxation" of the 0-Matrix in our theory.

### 0.5. The Birth of the Membrane (3D Self-Organization)

How did our ordered world emerge from 4-dimensional chaos? The answer is provided by wave mechanics and continuum physics. When fine sand is poured onto a vibrating membrane, the grains do not scatter

chaotically but form precise geometric patterns—known as Chladni figures. Similarly, under the influence of permanent pressure waves, 0-particles are compelled to self-organize into a gigantic, coherent network.

From the principles of spatial geometry, it follows that a stable phase boundary in a 4-dimensional medium possesses exactly 3 dimensions. Our Universe is precisely such an elastic, 3-dimensional Membrane, suspended within the depths of the 4th dimension. Everything we observe consists of vibrations propagating across this gigantic, oscillating "membrane." When this structure undergoes a local, non-linear distortion (as a result of exceeding the plasticity threshold of the matrix), a permanent topological braid is formed within it—a physical particle.

Since this distortion also occupies a volume in the fourth dimension, it causes a permanent bulging of our Membrane. This local curvature, resulting from the presence of the knot, creates a stress gradient in the surrounding structure. It is this geometric deformation of the Membrane toward the 4th dimension, induced by the pressure of the surrounding matrix, that we perceive macroscopically as gravity.

## 0.6. The Axiom of Physical Finiteness: Categorical Exclusion of Singularities and "Black Holes"

At this juncture, we must formulate a fundamental logical postulate that represents the deepest line of demarcation between TSM and the 20th-century legacy of General Relativity (GR) and standard quantum mechanics.

**The TSM theory introduces the Axiom of Physical Finiteness: in the real world, infinities do not exist.** Mathematical concepts in which any physical parameter—such as density, pressure, temperature, or spatial curvature—tends toward infinity ( $\infty$ ) are purely artifacts of imperfect, continuum-like mathematical models devoid of a mechanistic basis.

The consequence of the non-zero intrinsic volume of 0-particles ( $V_0 > 0$ ) and the existence of a geometric limit to their packing ( $\phi_{\max}$ ) is a categorical prohibition against any region of space collapsing into a dimensionless point. Because 0-particles possess hard kinetic cores, no force in the Universe—including the most extreme gravitational pressure—is capable of compressing matter below its structural geometric limit.

For this reason, the TSM model definitively and unambiguously excludes the existence of **gravitational singularities**, thereby refuting the concept of "black holes" in their classical, relativistic understanding (as objects of infinite density hidden behind an event horizon). Astronomical objects that modern astrophysics observes in galactic centers and erroneously interprets as black holes are, in reality, physical, volumetric structures of finite, extreme density. **In the architecture of TSM, we define these objects as degenerate, macroscopic matrix braids—plasmoids.** A plasmoid is an ultra-efficient, relativistic mechanical engine that is a natural consequence of the packing limit of 0-particles. Its precise hydrodynamic, electromagnetic, and topological nature will be described in detail in subsequent chapters of this work; however, the reader must accept this physical substitute now and reject the dogma of infinite gravitational collapse in order to properly analyze the mechanics of the matrix.

## 0.7. Creation of the Primary Distortions: An Alternative to the Big Bang

In our model, we must delineate a clear boundary of knowledge. The question of what imparted the initial momentum to the 0-particles and what triggered the first "distortions" within the 0-Matrix remains an open issue. Inquiring into the causes of the primary state of the system lies beyond the strict scope of this theory; it belongs to the realm of pure philosophical speculation rather than verifiable mechanics.

We can, however, relying on the Axiom of Physical Finiteness, formulate a rigorous alternative to the classical concept of the "Big Bang." Since the initial singularity phase is geometrically impossible due to the incompressibility of the 0-particles themselves, the TSM model rejects a singular, non-physical act of creation of everything from a point of zero volume.

Instead, the model allows for a scenario that is far more rational and consistent with continuum physics. Within our 4-dimensional vessel, countless local and large-scale pressure fluctuations have always occurred and continue to occur. A powerful, cumulative wave of kinetic energy, striking the already formed 3-dimensional Membrane, forces its rapid, local, non-linear "coiling." It is precisely in this manner, through the local exceeding of the matrix's plasticity thresholds and the mass generation of permanent topological braids, that the first clusters of matter in the Universe were formed and can continuously be created.